Taken together, IAM's approach to PARC reflects its domain-specific design: each element is defined and managed by AWS, enabling deep integration and precise control but limiting portability outside the AWS environment. For ACME, using IAM to secure S3 documents meant they could utilize AWS-native features—like principal and resource tags from Identity Center—to implement attribute-based access without creating a separate authorization layer.

8.5 Comparing policy languages

ACME's journey through different approaches to policy illustrates how the right authorization system depends on the problem you're addressing, the environment you're working in, and the skills of the team putting it into action. The following sections give a clear overview of each system using the PARC model as a perspective.

8.5.1 XACML

XACML was the first widely accepted standard for externalized authorization, with strong attribute-based logic, but uses a verbose XML syntax that limited developer adoption:

- Principal: Shown as attributes in the request; no first-class identity model.
- Action: Represented as attributes, usually strings like `"read"` or `"edit"`.
- Resource: Entirely described through attributes; no native object hierarchy or relationships.
- Context: Robust ABAC support with extensive environment attributes.
- Strengths: Uses standards, is expressive, and clearly separates policy from enforcement.
- Limitations: Verbose XML syntax, limited modern tools, heavyweight PDP/PEP architecture.

XACML's attribute model works well in compliance-focused environments but seems outdated in modern development.